



FINAL EXAMINATION

Course: Discrete Structure	Class: 22TT2
Time: 120 minutes	Term: 1 - Academic year: 2023-2024
Lecturer(s): Nguyễn Tấn Trung	
Student name:	Student ID:

(Notes: Books and phones are NOT allowed)

<Followings are the questions and/or requirements>

Question 1: (2 pt) Answer these questions for the poset

$$(\{\{1\}, \{2\}, \{4\}, \{1, 2\}, \{1, 4\}, \{2, 4\}, \{3, 4\}, \{1, 3, 4\}, \{2, 3, 4\}\}, \subseteq)$$

- a) (0.5 pt) Find the maximal and minimal elements.
- b) (0.5 pt) Find the greatest and the least elements if it exists?
- c) (0.5 pt) Find all upper bounds and the least upper bound (if it exists) of $\{\{2\}, \{4\}\}$
- d) (0.5 pt) Find all lower bounds and the greatest lower bound (if it exists) of

$$\{\{1, 3, 4\}, \{2, 3, 4\}\}$$

Question 2: (2 pt) How many ways are there to distribute five balls into three boxes if each box must have at least one ball in it if

- a) Both the balls and boxes are labeled?
- b) The balls are labeled, but the boxes are unlabeled?
- c) The balls are unlabeled, but the boxes are labeled?
- d) Both the balls and boxes are unlabeled?

Question 3: (1.5 pt) Find a minimal expansion as a Boolean sum of Boolean products of the function in the variables of w, x, y and z

$$wxyz + wxy\bar{z} + wx\bar{y}z + w\bar{x}\bar{y}z + w\bar{x}\bar{y}\bar{z} + \bar{w}x\bar{y}z + \bar{w}x\bar{y}\bar{z} + \bar{w}\bar{x}\bar{y}\bar{z}$$

- a) (0.5 pt) Use a K-map.
- b) (1 pt) Use the Quine-McCluskey method.

Question 4: (1.5 pt) A string that contains only 0s, 1s, and 2s is called a **ternary string**

- a) Find a recurrence relation for the number of **ternary strings** of length n that do not contain two consecutive 0s or two consecutive 1s.

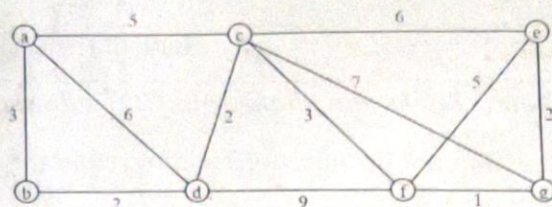


b) What are the initial conditions?

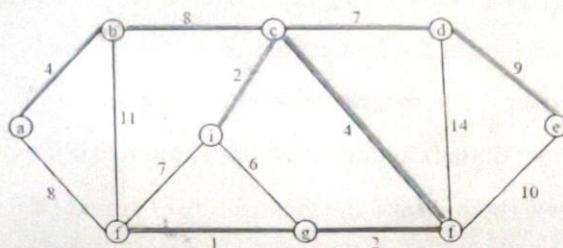
c) Solve this recurrence relation with the initial conditions found in the above sentence.

Question 5: (1 pt) Use *Dijkstra's algorithm* to find the shortest path between "a" and "g".

Explain your solution.



Question 6: (1.5 pt) Find a minimum spanning tree for the given weighted graph using



a) Prim's algorithm

b) Kruskal's algorithm

Question 7: (0.5 pt) Find all solutions, if any, to the system of congruences

$$x \equiv 7 \pmod{9}, \quad x \equiv 4 \pmod{12}, \quad \text{and} \quad x \equiv 16 \pmod{21}.$$

--- THE END ---